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- d calculate a value for the equilibrium constant for a reaction based on data from experiment, eg the reaction of ethanol and ethanoic acid (this can be used as an example of the use of ICT to present and analyse data), the equilibrium $\text{Fe}^{2+}(\text{aq}) + \text{Ag}^+(\text{aq}) \rightleftharpoons \text{Fe}^{3+}(\text{aq}) + \text{Ag}(\text{s})$ or the distribution of ammonia or iodine between two immiscible solvents
 - e construct expressions for K_c and K_p for homogeneous and heterogeneous systems, in terms of equilibrium concentrations or equilibrium partial pressures, perform simple calculations on K_c and K_p and work out the units of the equilibrium constants
 - f demonstrate an understanding that when ΔS_{total} increases the magnitude of the equilibrium constant increases since $\Delta S = R \ln K$
 - g apply knowledge of the value of equilibrium constants to predict the extent to which a reaction takes place
 - h relate the effect of a change in temperature on the value of ΔS_{total} .

4.6 Application of rates and equilibrium

Students will be assessed on their ability to:

- a demonstrate an understanding of how, if at all, and why a change in temperature, pressure or the presence of a catalyst affects the equilibrium constant and the equilibrium composition and recall the effects of changes of temperature and pressure on rate, eg the thermal decomposition of ammonium chloride, or the effect of temperature and pressure changes in the system $2\text{NO}_2 \rightleftharpoons \text{N}_2\text{O}_4$
- b use information on enthalpy change and entropy to justify the conditions used to obtain economic yields in industrial processes, and understand that in reality industrial processes cannot be in equilibrium since the products are removed, eg in the Haber process temperature affects the equilibrium yield and rate whereas pressure affects only the equilibrium yield (knowledge of industrial conditions are not required)
- c demonstrate an understanding of the steps taken in industry to maximise the atom economy of the process, eg recycling unreacted reagents or using an alternative reaction